Part A

Problem 1. We implement the steepest descent algorithm to minize the following function:

$$f(x) = 5x_1^2 + x_2^2 + 4x_1x_2 - 14x_1 - 6x_2 + 20$$

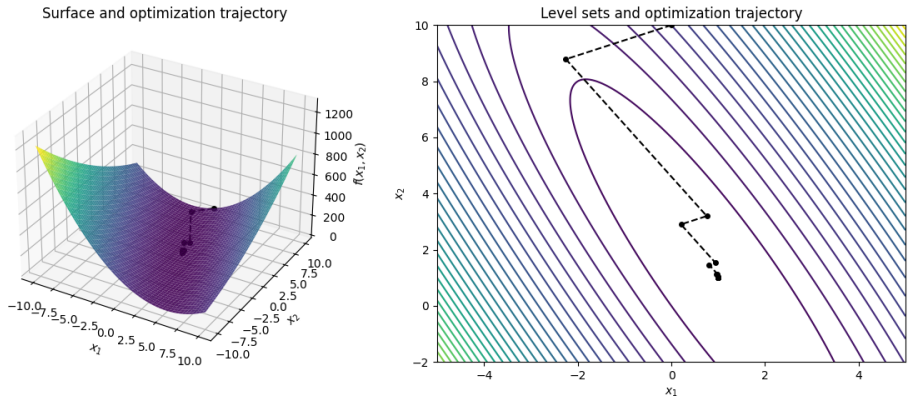
We write a python script to solve this problem using Steepest Descent algorithm.

```python
# f(x) = 5x1^2 + x2^2 + 4x1x2 - 14x1 - 6x2 + 20
def f(x):
    return ((5 * (x[0] ** 2)) + (x[1] ** 2) + (4 * x[0] * x[1]) - (14 * x[0]) - (6 * x[1]) + 20)

# ∇f(x) = [10x1 + 4x2 - 14, 2x2 + 4x1 - 6]
def grad_f(x):
    return np.array([10 * x[0] + 4 * x[1] - 14, 2 * x[1] + 4 * x[0] - 6])

# Find the stepsize ak that minimizes f(xk + ak * dk)
def find_stepsize(d):
    alpha = ((d[0] ** 2) + (d[1] ** 2)) / (2 * (5 * (d[0] ** 2) + (d[1] ** 2) + 4 * (d[0] * d[1])))
    return alpha

if __name__ == "__main__":
    x = np.array([0.0, 10.0])                   # start at x0 = (0,10)
    epsilon = 1e-6                              # ε : threshold

    k = 0
    print("k | x1k | x2k | d1k | d2k || norm(dk)2 | αk | f(xk) ")
    while True:
        d = -grad_f(x)                          # dk : descent direction
        norm_d = np.linalg.norm(d)
        if norm_d < epsilon:
            print("Converged: ||dk|| < ε")
            print(f"{k} | {xk[0]:.6f} | {xk[1]:.6f} | {d[0]:.6f} | {d[1]:.6f} | {norm_d:.6f} | {alpha:.4f} | {f(xk):.6f} ")
            break

        alpha = find_stepsize(d)                # αk : step size
        xk = x
        x = x + alpha * d                       # xk+1 = xk + αk * dk

        print(f"{k} | {xk[0]:.6f} | {xk[1]:.6f} | {d[0]:.6f} | {d[1]:.6f} | {norm_d:.6f} | {alpha:.4f} | {f(xk):.6f} ")
        k += 1
```

We use the results to build the following table,

k	x_1^k	x_2^k	d_1^k	d_2^k	$\|d^k\|_2$	α_k	$f(x^k)$
0	0.000000	10.000000	−26.000000	−14.000000	29.529646	0.0866	60.000000
1	−2.252782	8.786963	1.379968	−2.562798	2.910712	2.1800	22.222576
2	0.755548	3.200064	−6.355739	−3.422321	7.218567	0.0866	12.987827
3	0.204852	2.903535	0.337335	−0.626480	0.711528	2.1800	10.730379
4	0.940243	1.537809	−1.553670	−0.836592	1.764590	0.0866	10.178542
5	0.805625	1.465322	0.082462	−0.153144	0.173934	2.1800	10.043645
6	0.985392	1.131468	−0.379797	−0.204506	0.431357	0.0866	10.010669
7	0.952485	1.113749	0.020158	−0.037436	0.042518	2.1800	10.002608
8	0.996429	1.032138	−0.092842	−0.049992	0.105446	0.0866	10.000638
9	0.988385	1.027806	0.004928	−0.009151	0.010394	2.1800	10.000156
10	0.999127	1.007856	−0.022695	−0.012221	0.025776	0.0866	10.000038
11	0.997161	1.006797	0.001205	−0.002237	0.002541	2.1800	10.000009
12	0.999787	1.001920	−0.005548	−0.002987	0.006301	0.0866	10.000002
13	0.999306	1.001662	0.000294	−0.000547	0.000621	2.1800	10.000001
14	0.999948	1.000469	−0.001356	−0.000730	0.001540	0.0866	10.000000
15	0.999830	1.000406	0.000072	−0.000134	0.000152	2.1800	10.000000
16	0.999987	1.000115	−0.000332	−0.000179	0.000377	0.0866	10.000000
17	0.999959	1.000099	0.000018	−0.000033	0.000037	2.1800	10.000000
18	0.999997	1.000028	−0.000081	−0.000044	0.000092	0.0866	10.000000
19	0.999990	1.000024	0.000004	−0.000008	0.000009	2.1800	10.000000
20	0.999999	1.000007	−0.000020	−0.000011	0.000023	0.0866	10.000000
21	0.999998	1.000006	0.000001	−0.000002	0.000002	2.1800	10.000000
22	1.000000	1.000002	−0.000005	−0.000003	0.000006	0.0866	10.000000
Converged: $\|d^k\| < \varepsilon$							
23	1.000000	1.000002	0.000000	−0.000000	0.000001	0.0866	10.000000

To visualize the algorithm we plot the function, level sets, and the algorithm trajectory.



Problem 2. We use the a modified version of the algorithm in **Problem 1.** for the following function,

$$f(x) = \frac{1}{2}x^\top Qx + C^\top x + 10,$$

(i) Under the following conditions, $x0 = \begin{bmatrix} 40 \\ -100 \end{bmatrix}, \quad Q = \begin{bmatrix} 20 & 5 \\ 5 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 14 \\ 6 \end{bmatrix}$

k	x_1^k	x_2^k	d_1^k	d_2^k	$\|d^k\|_2$	α_k	$f(x^k)$
0	40.000000	-100.000000	−314.000000	−6.000000	314.057320	0.049543	5970.000000
1	24.443465	-100.297259	−1.383004	72.377193	72.390405	0.550795	3526.732204
2	23.681713	-60.432247	−185.473024	−3.544071	185.506882	0.049543	2083.546642
3	14.492803	-60.607832	−0.816910	42.751646	42.759450	0.550795	1231.088073
4	14.042853	-37.060430	−109.554913	−2.093406	109.574912	0.049543	727.559192
5	8.615162	-37.164144	−0.482531	25.252475	25.257085	0.550795	430.135548
6	8.349386	-23.255201	−64.711723	−1.236530	64.723536	0.049543	254.453820
7	5.143367	-23.316463	−0.285021	14.916092	14.918815	0.550795	150.682414
8	4.986379	-15.100751	−38.223818	−0.730391	38.230796	0.049543	89.386884
9	3.092652	-15.136937	−0.168356	8.810614	8.812222	0.550795	53.180937
⋮				...			
62	0.133337	-3.333341	−0.000026	−0.000000	0.000026	0.049543	0.933333
63	0.133335	-3.333341	−0.000000	0.000006	0.000006	0.550795	0.933333
64	0.133335	-3.333338	−0.000015	−0.000000	0.000015	0.049543	0.933333
65	0.133335	-3.333338	−0.000000	0.000003	0.000003	0.550795	0.933333
66	0.133334	-3.333336	−0.000009	−0.000000	0.000009	0.049543	0.933333
67	0.133334	-3.333336	−0.000000	0.000002	0.000002	0.550795	0.933333
68	0.133334	-3.333335	−0.000005	−0.000000	0.000005	0.049543	0.933333
69	0.133334	-3.333335	−0.000000	0.000001	0.000001	0.550795	0.933333
70	0.133334	-3.333334	−0.000003	−0.000000	0.000003	0.049543	0.933333
Converged: $\|d^k\| < \varepsilon$							
71	0.133334	-3.333334	−0.000000	0.000001	0.000001	–	0.933333

(ii) Under the following conditions, $x0 = \begin{bmatrix} 40 \\ -100 \end{bmatrix}$, $\quad Q = \begin{bmatrix} 20 & 5 \\ 5 & 16 \end{bmatrix}$, $\quad C = \begin{bmatrix} 14 \\ 6 \end{bmatrix}$

k	x_1^k	x_2^k	d_1^k	d_2^k	$\|d^k\|_2$	α_k	$f(x^k)$
0	40.000000	-100.000000	-314.000000	1394.000000	1428.926870	0.071177	75970.000000
1	17.650308	-0.778755	-363.112381	-81.791454	372.210214	0.045557	3303.890780
2	1.108047	-4.504917	-13.636350	60.538444	62.055244	0.071177	148.156778
3	0.137447	-0.195951	-15.769196	-3.552028	16.164295	0.045557	11.109987
4	-0.580948	-0.357770	-0.592198	2.629055	2.694927	0.071177	5.158337
5	-0.623099	-0.170641	-0.684823	-0.154257	0.701981	0.045557	4.899870
6	-0.654297	-0.177668	-0.025718	0.114174	0.117035	0.071177	4.888645
7	-0.656128	-0.169541	-0.029740	-0.006699	0.030486	0.045557	4.888158
8	-0.657483	-0.169847	-0.001117	0.004958	0.005083	0.071177	4.888137
9	-0.657562	-0.169494	-0.001292	-0.000291	0.001324	0.045557	4.888136
10	-0.657621	-0.169507	-0.000049	0.000215	0.000221	0.071177	4.888136
11	-0.657624	-0.169492	-0.000056	-0.000013	0.000057	0.045557	4.888136
12	-0.657627	-0.169492	-0.000002	0.000009	0.000010	0.071177	4.888136
13	-0.657627	-0.169492	-0.000002	-0.000001	0.000002	0.045557	4.888136
Converged: $\|d^k\| < \varepsilon$							
14	-0.657627	-0.169492	-0.000000	0.000000	0.000000	–	4.888136

Part B

Problem 1. Suppose that $f : \mathbb{R}^3 \to \mathbb{R}^2$ is defined by

$$f(x, y, z) = \begin{bmatrix} x^2 + yz \\ sin(xyz) + z \end{bmatrix}$$

(i) Compute the Jacobian of f at $(x, y, z) = (-1, 0, 1)$.

We compute the Jacobian of f,

$$J(x) = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \end{bmatrix} = \begin{bmatrix} 2x & z & y \\ yz\cos(xyz) & xz\cos(xyz) & xy\cos(xyz) + 1 \end{bmatrix}$$

Evaluating at $(x, y, z) = (-1, 0, 1)$,

$$J(-1, 0, 1) = \begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

(ii) Are there any directions in which the directional derivative of f at $(-1, 0, 1)$ zero? If yes, find them.

$$\begin{bmatrix} -2 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} -2d_1 + d_2 \\ -d_2 + d_3 \end{bmatrix} = 0$$

Then, $d_2 = 2d_1$ and $d_2 = d_3$ if $d_1 = 1$, then $d_2 = d_3 = 2$. So

$$d = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix},$$

is a direction.

Problem 2. Consider the function $f(x) = 2x_1^2 + x_1x_2 + x_2^2 + x_2x_3 + x_3^2 - 6x_1 - 7x_2 - 8x_3 + 9$
(i) Compute the gradient and Hessian of the function f.

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \frac{\partial f}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 4x_1 + x_2 - 6 \\ x_1 + 2x_2 + x_3 - 7 \\ x_2 + 2x_3 - 8 \end{bmatrix}$$

The Hessian is,

$$\nabla^2 f(x) = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

(ii) Evaluate the gradient and Hessian of f at $(0, 0, 0)$.

$$\nabla f((0,0,0)) = \begin{bmatrix} -6 \\ -7 \\ -8 \end{bmatrix}$$

$$\nabla^2 f((0,0,0)) = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

(iii) Obtain the second order approximation of f at $x^* = (0, 0, 0)$, i.e., the Taylors' series up to the second order.

$$f(x) \approx f(x^*) + \nabla f(x^*)^T (x - x^*) + \frac{1}{2}(x - x^*)^T \nabla^2 f(x^*)(x - x^*).$$

At $x^* = (0, 0, 0)$,

$$f(x) \approx 9 + \begin{bmatrix} -6 & -7 & -8 \end{bmatrix} x + \frac{1}{2} x^T \begin{bmatrix} 4 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} x.$$

Problem 3. Let $A \in \mathbb{R}^{n \times n}$ and $x \in \mathbb{R}^n$. Compute $\frac{dy}{dx}$ for the following.
Not Answered.

Problem 4. Suppose that the temperature at each point of a metal plate is given by

$$T(x, y) = e^x \cos y + e^y \cos x$$

(i) In what direction does the temperature increase most rapidly at point $(0, 0)$? and what is the rate of increase?

The gradient is

$$\nabla T(x, y) = \begin{bmatrix} e^x \cos y - e^y \sin x \\ -e^x \sin y + e^y \cos x \end{bmatrix}.$$

Let $x^* = (0, 0)$ then at x^*,

$$\nabla T(x^*) = \begin{bmatrix} \cos(0) - \sin(0) \\ -\sin(0) + \cos(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The temperature increases most rapidly in the direction of $\nabla T(x^*) = \begin{bmatrix} 1 & 1 \end{bmatrix}^\top$. The maximum rate of increase is

$$\|\nabla T(x^*)\| = \sqrt{2}.$$

(ii) In what direction does the temperature decrease most rapidly at point $(0, 0)$?

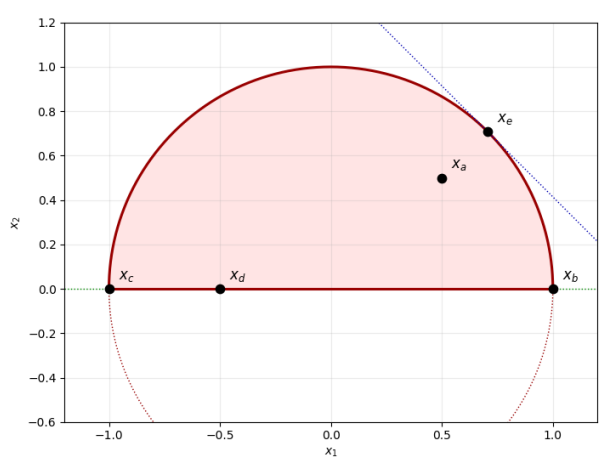
If the temperature increases most rapidly at $\nabla T(x^*) = \begin{bmatrix} 1 & 1 \end{bmatrix}^\top$, then the tempereature decreases most rapidly at $-\nabla T(x^*) = \begin{bmatrix} -1 & -1 \end{bmatrix}^\top$.

Problem 5. Consider the feasible region defined by the constraints

$$\begin{aligned} 1 - x_1^2 - x_2^2 &\geq 0 \\ \sqrt{2} - x_1 - x_2 &\geq 0 \\ x_2 &\geq 0 \end{aligned}$$

For each of the following points, check if it is in the feasible region or not. If it is in the feasible region, check whether the point is in the interior or on the boundary of each of the constraints.

We see that all points $x_a = (\frac{1}{2}, \frac{1}{2})^\top$, $x_b = (1, 0)^\top$, $x_c = (-1, 0)^\top$ $x_d = (-\frac{1}{2}, 0)^\top$, $x_e = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})^\top$, are in the feasible region. With the point x_a, being in the interior, and all other points, x_b, x_c, x_d, x_e are on the boundary of the feasible region.



Problem 6. In each of the following problems fully justify your answer using optimality conditions.

(i) Show that $f(x,y) = (x^2-4)^2 + y^2$ has two global minima and one stationary point, which is neither a local maximum nor a local minimum.

First we compute $\nabla f(x) = \begin{bmatrix} 4x^3 - 16x \\ 2y \end{bmatrix}$, then we solve $\nabla f(x) = 0$. It is easy to see that $y = 0$, so now we solve for x. We have $4x^3 - 16x = x^3 - 4x = x(x^2-4) = x(x-2)(x+2) = 0$ when $x = 0, 2,$ or -2.

For the stationaty point $(0,0)$, we compute $\nabla^2 f(x) = \begin{bmatrix} 12x^2 - 16 & 0 \\ 0 & 2 \end{bmatrix}$, then $\nabla^2 f((0,0) = \begin{bmatrix} -16 & 0 \\ 0 & 2 \end{bmatrix}$.

Since, $\det(\begin{bmatrix} -16 & 0 \\ 0 & 2 \end{bmatrix}) = -32 < 0$, then $(0,0)$ is a saddle point at $f(x,y) = 16$.

For the stationary points $(2,0)$ and $(-2,0)$, $\nabla^2 f((\pm 2, 0) = \begin{bmatrix} 32 & 0 \\ 0 & 2 \end{bmatrix}$, and $\det(\begin{bmatrix} 32 & 0 \\ 0 & 2 \end{bmatrix}) = 64 > 0$, thus $(2,0)$ and $(-2,0)$ are minima.

Notice, since $(x^2-4)^2 \geq 0$ and $y^2 \geq 0$, then at $f(x,y) = 0$,

$$0 = (x^2-4)^2 + y^2 \implies (x^2-4)^2 = 0 \text{ and } y^2 = 0 \implies f(x,y) \geq 0, \text{ for all } x, y.$$

Thus, the points $(2,0)$ and $(-2,0)$ are both global minimum at $f(x,y) = 0$.

(ii) Find all local minima of $f(x,y) = 1/2x^2 + x\cos(y)$.

First we compute $\nabla f(x) = \begin{bmatrix} x + \cos(y) \\ -x\sin(y) \end{bmatrix}$, and solve for $\nabla f(x) = 0$.

From the second component $-x\sin(y) = 0$, by the zero product property, either $x = 0$ or $\sin(y) = 0$.

If $x = 0$, then $x + \cos(y) = 0 \implies \cos(y) = 0$, so $y = \frac{\pi}{2} + k\pi$ for all $k \in \mathbb{Z}$. Thus the following are all stationary points,

$$(0, \frac{\pi}{2} + k\pi), \quad \forall k \in \mathbb{Z}$$

If $\sin(y) = 0$, then $y = k\pi$ for all $k \in \mathbb{Z}$. So $x + \cos(y) = 0 \implies x = -\cos(y)$. Since $y = k\pi$, then if k is even, $x = -\cos((2k)\pi) = -(1) = -1$ and if k is odd, then $x = -\cos((2k+1)\pi) = -(-1) = 1$. Thus the following are all stationary points,

$$(-1, 2k\pi) \quad \text{and} \quad (1, (2k+1)\pi)$$

Next we compute the Hessian, $\nabla^2 f(x) = \begin{bmatrix} 1 & -\sin(y) \\ -\sin(y) & -x\cos(y) \end{bmatrix}$.

At points $(0, \frac{\pi}{2} + k\pi)$,

$$\nabla^2 f = \begin{bmatrix} 1 & \pm 1 \\ \pm 1 & 0 \end{bmatrix} \quad \implies \quad \det(\begin{bmatrix} 1 & \pm 1 \\ \pm 1 & 0 \end{bmatrix}) = -1 < 0,$$

Thus, these are saddle points.

At points $(-1, 2k\pi)$ and $(1, (2k+1)\pi)$

$$\nabla^2 f = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \Longrightarrow \quad \det(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}) = 1 > 0,$$

Since $\nabla^2 f$ is positive definite all points $(-1, 2k\pi)$ and $(1, (2k+1)\pi)$ are strict local minima.

(iii) Find all local minima and all local maxima of $f(x, y) = \sin x + \sin y + \sin(x + y)$ within the set $\{(x, y)|0 < x < 2\pi, 0 < y < 2\pi\}$.

First we compute $\nabla f(x) = \begin{bmatrix} \cos(x) + \cos(x+y) \\ \cos(y) + \cos(x+y) \end{bmatrix}$, and solve for $\nabla f(x) = 0$.

From the first component $\cos(x) + \cos(x+y) = 0 \implies \cos(x+y) = -\cos(x)$. Substituting into the second component, $\cos(y) - \cos(x) = 0 \implies \cos(y) = \cos(x)$. Thus, $x = y$, so $\cos(x) + \cos(2x) = 0$ in the domain $0 < x < 2\pi$, the solutions are $\cos(\frac{\pi}{3}) + \cos(\frac{2\pi}{3}) = \frac{1}{2} - \frac{1}{2} = 0$, $\cos(\pi) + \cos(2\pi) = -1 + 1 = 0$, and $\cos(\frac{5\pi}{3}) + \cos(\frac{10\pi}{3}) = \frac{1}{2} - \frac{1}{2} = 0$. We know $x = y$, so the critical points are $(\frac{\pi}{3}, \frac{\pi}{3}), (\pi, \pi)$, and $(\frac{5\pi}{3}, \frac{5\pi}{3})$.

Next we compute the Hessian. $\nabla^2 f(x) = \begin{bmatrix} -\sin(x) - \sin(x+y) & -\sin(x+y) \\ -\sin(x+y) & -\sin(y) - \sin(x+y) \end{bmatrix}$.

At $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$,

$$\nabla^2 f = \begin{bmatrix} -\sqrt{3} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\sqrt{3} \end{bmatrix}, \quad \text{and} \quad \det(\begin{bmatrix} -\sqrt{3} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\sqrt{3} \end{bmatrix}) = 2\sqrt{3} - \sqrt{3} > 0.$$

which is negative definite, thus $\left(\frac{\pi}{3}, \frac{\pi}{3}\right)$ is a strict local maximum.
Similarly for $\left(\frac{5\pi}{3}, \frac{5\pi}{3}\right)$,

$$\nabla^2 f = \begin{bmatrix} \sqrt{3} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \sqrt{3} \end{bmatrix}, \quad \text{and} \quad \det(\begin{bmatrix} \sqrt{3} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \sqrt{3} \end{bmatrix}) = 2\sqrt{3} - \sqrt{3} > 0.$$

but is positive definite, thus $\left(\frac{5\pi}{3}, \frac{5\pi}{3}\right)$ is a strict local minimum.
Then, at (π, π),

$$\nabla^2 f = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Thus, this point is niether a minimum or a maximum.

Problem 7. Find all the descent directions of the following function at the point $(0, 0)^T$, $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$.

First we compute $\nabla f(x) = \begin{bmatrix} -400x_1x_2 + 400x_1^3 - 2 + 2x_1 \\ 200x_2 - 200x_1^2 \end{bmatrix}$.

At $(0, 0)$,

$$\nabla f(0, 0) = \begin{bmatrix} -2 \\ 0 \end{bmatrix}.$$

We want $\nabla f(0,0)^T d < 0$,

$$\nabla f(0,0)^T d = \begin{bmatrix} -2 & 0 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = -2d_1.$$

Thus, if $-2d_1 < 0$, then $d_1 > 0$. Therefore, all descent directions at $(0,0)$ are $d = (d_1, d_2)^T$ with $d_1 > 0$.

Problem 8.

a. Give an example for an optimization problem in your own words. Try to describe a problem that has real implications.

An NFL team needs to sign a player to fill their roster needs. Each free agent has a contract cost and estimated performance values. The team wants the best overall player it can afford.

This has real implications for the sports industry, but it could also be formulated as any company hiring a new employee.

b. Identify all the information relevant to formulate meaningful constraints.

Each player in the free agent set, has a contract cost and a set of performancee attributes on a scale from 1 to 10. The team must select the highest performing player while minimizing the difference of their contract to the alloted budget, where going over and under is allowed.

The primary constraint is the team's budget for signing that position. Then, the other constraints would be the set of available players for the target position. The cost of each player's contract, and some minimum thresholds for the performance values.

c. Formulate/Derive the mathematical model of the stated optimization problem.

Suppose there are m free agents indexed by $i = 1, \ldots, m$. We have scouting data for each free agent that provides K attributes, denoted $a_{i,k} \in \{1, \ldots, 10\}$ for $k = 1, \ldots, K$. Let c_i be the contract cost of player i, and b is the target budget. Then $d_i = |c_i - b|$ is the contract difference from the target budget for that player. We set a weight parameter $\lambda > 0$ to for tuning the penalty for deviating from the target budget.

$$\begin{aligned}
\max_{p,d} \quad & \sum_{i=1}^{m} g(a_i)\, p_i \; - \; \lambda d_i \\
\text{s.t.} \quad & g(a_i) = \sum_{k=1}^{K} a_{i,k}, \\
& \sum_{i=1}^{m} p_i = 1, \qquad p_i \in \{0,1\}, \; \forall i, \\
& -d \leq \sum_{i=1}^{m} c_i p_i - b \leq d, \\
& d \geq 0, \qquad b, c > 0, \qquad 1 \leq a_{i,k} \leq 10.
\end{aligned}$$

Appendix

A1

All files and code can be found on Github.

`https://github.com/allanpaiz/csce-590/tree/main/hw3`

Sources

1. **Use of AI** During the completion of this assignment, I used ChatGPT 2.5 (OpenAI) as an assistive tool. I reviewed and edited all AI assisted outputs, and take full responsibility for the accuracy of my work.
2. Lecture notes from **CSCE 590** : *Optimization* by Narayanan, Vignesh.
3. Lecture notes from **MATH 528** : *Math Foundations of ML and DS* by Colebank, Mitchel.
4. Homework 3 code from **MATH 528** : *Math Foundations of ML and DS* by Colebank, Mitchel.